

SF1811 Optimization

Lecture 5: Duality and optimality in linear programming

Chapter 6 and the note on LP optimality and duality

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Duality and optimality in linear programming

Main question:

How can we certify that a feasible solution is optimal?

Today:

- Simplex multipliers and reduced costs.
- The dual linear program.
- Weak and strong duality.
- Complementarity and optimality conditions.
- Degeneracy and perturbation.

What does the simplex method give us?

Consider the primal problem in standard form:

$$(P_{\mathrm{LP}}) \qquad \begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & Ax = b, \\ & x \geq 0. \end{array}$$

For a current basic feasible solution,

$$A_B x_B = b, \qquad x_B \geq 0, \qquad x_N = 0.$$

The simplex multipliers and reduced costs satisfy

$$A_B^{\mathsf{T}} y = c_B, \qquad r_N = c_N - A_N^{\mathsf{T}} y.$$

The simplex method terminates when

$$r_N \geq 0.$$

What does this stopping condition certify?

The full reduced-cost vector

Define

$$r = c - A^{\mathsf{T}} y.$$

Partitioning according to the basic and nonbasic tuples gives

$$r_B = c_B - A_B^{\mathsf{T}} y, \qquad r_N = c_N - A_N^{\mathsf{T}} y.$$

Since

$$A_B^{\mathsf{T}} y = c_B,$$

we obtain

$$r_B = 0.$$

Hence, at termination,

$$r_B = 0, \qquad r_N \geq 0, \qquad r \geq 0.$$

The dual linear program

For the primal problem

$$(P_{\mathrm{LP}}) \qquad \begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & Ax = b, \\ & x \geq 0, \end{array}$$

the simplex stopping conditions suggest the problem

$$(D_{\mathrm{LP}}) \qquad \begin{array}{ll} \text{maximize} & b^{\mathsf{T}} y \\ \text{subject to} & A^{\mathsf{T}} y + r = c, \\ & r \geq 0. \end{array}$$

The variable $y \in \mathbb{R}^m$ is free. The variable $r \in \mathbb{R}^n$ is nonnegative.

Equivalently, eliminating r gives

$$\begin{array}{ll} \text{maximize} & b^{\mathsf{T}} y \\ \text{subject to} & A^{\mathsf{T}} y \leq c. \end{array}$$

Why is $b^{\mathsf{T}} y$ a lower bound on the primal objective value?

Weak duality

Let x be feasible to (P_{LP}) and let (y, r) be feasible to (D_{LP}). Then

$$\begin{aligned} c^{\mathsf{T}}x - b^{\mathsf{T}}y &= (A^{\mathsf{T}}y + r)^{\mathsf{T}}x - b^{\mathsf{T}}y \\ &= y^{\mathsf{T}}Ax + r^{\mathsf{T}}x - b^{\mathsf{T}}y \\ &= r^{\mathsf{T}}x \\ &\geq 0. \end{aligned}$$

Theorem (Weak duality for linear programming)

For every primal feasible x and every dual feasible (y, r),

$$b^{\mathsf{T}}y \leq c^{\mathsf{T}}x.$$

Consequently,

$$\text{optval}(D_{\text{LP}}) \leq \text{optval}(P_{\text{LP}}).$$

An optimality certificate

Suppose that x is feasible to (P_{LP}) and (y, r) is feasible to (D_{LP}).
If $c^{\mathsf{T}}x = b^{\mathsf{T}}y$, then x and y, r are primal and dual optimal solutions, respectively.

Proposition

If a primal feasible solution and a dual feasible solution have equal objective values, then both solutions are optimal.

Proof

Weak duality gives $b^{\mathsf{T}}y \leq \mathrm{optval}(D_{\mathrm{LP}}) \leq \mathrm{optval}(P_{\mathrm{LP}}) \leq c^{\mathsf{T}}x$.
If $c^{\mathsf{T}}x = b^{\mathsf{T}}y$, both inequalities are equalities. Hence x is primal optimal and (y, r) is dual optimal.

Complementarity

Since

$$x \geq 0, \qquad r \geq 0,$$

we have

$$r^{\mathsf{T}} x = 0 \quad \Longleftrightarrow \quad x_j r_j = 0, \qquad j = 1, \ldots, n.$$

Thus, for every j,

$$x_j > 0 \quad \Longrightarrow \quad r_j = 0,$$

and

$$r_j > 0 \quad \Longrightarrow \quad x_j = 0.$$

A primal variable and its corresponding dual slack cannot both be positive.

Complementarity in the simplex method

For a basic feasible solution,

$$x_N = 0.$$

For the simplex multipliers,

$$r_B = 0.$$

Consequently,

$$\begin{aligned} r^{\mathsf{T}} x &= r_B^{\mathsf{T}} x_B + r_N^{\mathsf{T}} x_N \\ &= 0. \end{aligned}$$

During the simplex method:

- x is primal feasible;
- $A^{\mathsf{T}} y + r = c$;
- $r^{\mathsf{T}} x = 0$;
- the missing condition is $r_N \geq 0$.

When $r_N \geq 0$, all primal-dual optimality conditions hold.

Strong duality: the nondegenerate case

Suppose that x is a nondegenerate optimal basic feasible solution:

$$x_B > 0, \qquad x_N = 0.$$

For each nonbasic variable x_k, sufficiently small positive values of x_k preserve feasibility after the basic variables are adjusted.

Optimality therefore implies

$$r_k \geq 0, \qquad k \in N.$$

Together with

$$r_B = 0,$$

we obtain a dual feasible solution (y, r) satisfying

$$c^\mathsf{T} x = b^\mathsf{T} y.$$

Hence x and (y, r) are primal and dual optimal solutions.

Degeneracy and perturbation

At a degenerate optimal basic feasible solution, not every associated basis need give nonnegative reduced costs.

A classical perturbation replaces

$$x_j \geq 0$$

by

$$x_j \geq -\epsilon^j, \qquad j = 1, \ldots, n.$$

For positive ϵ sufficiently close to zero, the perturbed problem is nondegenerate. An optimal basis of the perturbed problem then gives primal and dual optimal solutions to the original problem.

Degeneracy complicates the proof, but not the optimality conditions.

Strong duality for linear programming

$$(P_{\mathrm{LP}}) \qquad \begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & Ax = b, \\ & x \geq 0, \end{array} \qquad\qquad (D_{\mathrm{LP}}) \qquad \begin{array}{ll} \text{maximize} & b^{\mathsf{T}} y \\ \text{subject to} & A^{\mathsf{T}} y + r = c, \\ & r \geq 0. \end{array}$$

Theorem (Strong duality for linear programming)

If the primal problem has a finite optimal value, then the primal and dual problems have optimal solutions, and their optimal values are equal.

Thus there exist $\hat{x}$, $\hat{y}$, and $\hat{r}$ such that

$$c^{\mathsf{T}} \hat{x} = b^{\mathsf{T}} \hat{y}.$$

Linear programming optimality conditions

Corollary (Linear programming optimality conditions)

A vector $\hat{x}$ is an optimal solution to (P_{LP}) and $(\hat{y}, \hat{r})$ is an optimal solution to (D_{LP}) if and only if

1. $A\hat{x} = b, \quad \hat{x} \geq 0,$ *(Primal feasibility)*
2. $A^{\mathsf{T}}\hat{y} + \hat{r} = c, \quad \hat{r} \geq 0,$ *(Dual feasibility)*
3. $\hat{x}_j \hat{r}_j = 0, \quad j = 1, \ldots, n.$ *(Complementarity)*

The simplex stopping condition is a primal-dual optimality certificate.

Main ideas

After this lecture, you should be able to:

- formulate the dual of a linear program in standard form;
- prove weak duality;
- use equal primal and dual objective values to certify optimality;
- state and interpret the complementarity conditions;
- relate simplex multipliers and reduced costs to dual variables;
- explain the role of perturbation in the proof of strong duality.

Next lecture:

How does Lagrangian relaxation generate the dual problem and explain the meaning of the multipliers?